Effective dimensionality $d_{\text{eff}}(x, \tau)$ from MGD's fitted θ_t

Generated with Claude Code from code-reading/debugging sessions, 2026-09-23/24;
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Part I (Guth's d_{eff}, MGD equations, implementation) added 2026-10-15

Purpose

Guth et al. define a local effective dimensionality $d_{\text{eff}}(x, \tau)$ of a distribution's support, at a point x and noise scale τ, from the score of the variance-exploding (VE) noised distribution at x. This note derives d_{eff} directly in terms of MGD's own fitted quantities – θ_t, φ, $\nabla\varphi$ – so it can be computed as pure post-processing of an already-completed MGD run, with no new sampling/fitting. Implementation: `codes/effective_dimension.py`. Notebook (CSV loader for plots): `turbulence/effective_dimension_deff_final.ipynb`; the per-seed CSVs it reads are produced on the cluster by `deff_diagnostics/deff_latest_run_check.py`.

Headline conclusion (used in Section 11). The *denoiser form* (Eq. 27) is the right estimator to use in the paper. The *energy form* (Eq. 20) is strictly equivalent to it only when the model density $p_\theta(\cdot, \tau)$ satisfies the Fokker-Planck equation of the noising, which the moment-matched family produced by MGD does not. The measured energy–denoiser gap is a direct, quantitative read-out of this mismatch (Section 9), small at the trivial ends ($\tau \to 0, \infty$) and largest where the physics is doing the most work.

Reading order. Part I (Sections 1–3, added 2026-10-15) is a self-contained reference: (A) what d_{eff} is in Guth et al., (B) the MGD equations needed to read the code, (C) what `codes/effective_dimension.py` computes, line by line. Part II (Sections 4 onwards) is the original derivation/debugging record, kept for provenance.

Notation clash. Guth et al. write t for the *noise variance*. MGD writes $t \in [0, 1]$ for the *interpolant time*. Here the noise variance is always τ and t is always MGD time. Guth's equations are rewritten with $t \to \tau$.

Part I: reference

1 A. Effective dimensionality in Guth, Kadkhodaie & Simoncelli

Source: *Learning normalized image densities via dual score matching*, arXiv:2506.05310 (copy in `turbulence/guth_et_al_paper.pdf`), Sec. 3.3, App. A.2, App. B.3.

A.1 Setting

Data $x \sim p \in \mathbb{R}^d$. Variance-exploding noising: $y = x + \sqrt{\tau}\,\xi$, $\xi \sim \mathcal{N}(0, I_d)$, $\tau \in (0, \infty)$. Noisy marginal and energy (their Eq. 1):

$$q_\tau(y) = \int p(x)\,\mathcal{N}(y; x, \tau I_d)\,dx, \qquad U(y, \tau) = -\log q_\tau(y). \tag{1}$$

Space score via Miyasawa–Tweedie (their Eq. 2):

$$\nabla_y U(y, \tau) = \mathbb{E}\big[\tfrac{y-x}{\tau} \,\big|\, y\big] \quad \Longleftrightarrow \quad \mathbb{E}[x \mid y] = y - \tau \nabla_y U(y, \tau) = y + \tau \nabla_y \log q_\tau(y). \tag{2}$$

Because $q_\tau = p * \mathcal{N}(0, \tau I)$, it solves the heat equation $\partial_\tau q_\tau = \frac{1}{2}\Delta_y q_\tau$. Written for U, this is a Hamilton–Jacobi equation (their Eqs. 32–35):

$$\partial_\tau U = \tfrac{1}{2}\Delta_y U - \tfrac{1}{2}\|\nabla_y U\|^2. \tag{3}$$

A.2 Definition

Fix a clean point x and a scale τ. The two definitions are their Eq. 11 (denoiser form) and Eq. 12 (energy form):

$$d_{\text{eff}}^{\text{den}}(x, \tau) = \frac{1}{\tau}\,\mathbb{E}_\xi\big[\|x - \mathbb{E}[x \mid y]\|^2\big], \qquad y = x + \sqrt{\tau}\,\xi, \tag{4}$$

$$d_{\text{eff}}^{\text{en}}(x, \tau) = d - \partial_s\,\mathbb{E}_\xi\big[U(x + \sqrt{\tau}\,\xi, \tau)\big], \qquad s = \tfrac{1}{2}\log\tau, \quad \partial_s = 2\tau\partial_\tau. \tag{5}$$

$\mathbb{E}_\xi[\cdot]$ is over the noise only. x stays fixed.

Interpretation (their Sec. 3.3). If, at scale τ, the support of p near x looks like a k-dimensional affine subspace, then the optimal denoiser projects y onto it. The noise components along the subspace are kept, so the error is τk and $d_{\text{eff}} = k$. Equivalently, extra noise spreads mass only in the $d - k$ normal directions, so $\mathbb{E}[U \mid x] \simeq (d - k)\,\frac{1}{2}\log\tau$ and Eq. 5 returns k. Limits: $d_{\text{eff}} \to 0$ as $\tau \to \infty$ (compact support looks like a point); $d_{\text{eff}} \to d$ as $\tau \to 0$ for any density that is continuous and positive at x. Both definitions are dimensionless, not nats. $d_{\text{eff}} \geq 0$ always. Averaged over $x \sim p$, $\mathbb{E}_x d_{\text{eff}} \leq d$, since the MMSE is at most τd. Pointwise it can exceed d: in the Gaussian example below, a coordinate with $x_i^2 > 2\sigma_i^2 + \tau$ contributes more than 1.

Gaussian check. For $p = \mathcal{N}(0, \operatorname{diag}\sigma_i^2)$: $d_{\text{eff}}(x, \tau) = \sum_i (\tau x_i^2 + \sigma_i^4)/(\sigma_i^2 + \tau)^2$ (this note, Eq. 33). Each coordinate contributes ≈ 1 if $\sigma_i^2 \gg \tau$ and ≈ 0 if $\sigma_i^2 \ll \tau$.

A.3 Why the two forms agree, and when they do not

Two exact identities, taken from their App. B.3 and written so that the hypotheses are explicit:

1. *Tweedie + Stein (holds for any smooth U, with no assumption on where it comes from).* Insert $\mathbb{E}[x|y] = y - \tau\nabla U$ and apply Stein's lemma $\mathbb{E}_\xi[\xi \cdot g(x + \sqrt{\tau}\xi)] = \sqrt{\tau}\,\mathbb{E}_\xi[\nabla \cdot g]$:

$$d_{\text{eff}}^{\text{den}} = d + \tau\,\mathbb{E}_\xi\big[\|\nabla_y U\|^2 - 2\Delta_y U\big]. \tag{6}$$

2. *Energy derivative (uses Eq. 3).* Differentiate under the Gaussian $\mathcal{N}(y; x, \tau I)$, which itself solves the heat equation, and integrate by parts: $\partial_\tau \mathbb{E}_\xi[U] = \mathbb{E}_\xi[\frac{1}{2}\Delta U + \partial_\tau U]$. Define the *Hamilton–Jacobi residual* of whatever U is used:

$$R(y, \tau) := \partial_\tau U - \tfrac{1}{2}\Delta_y U + \tfrac{1}{2}\|\nabla_y U\|^2. \tag{7}$$

Then $\partial_\tau \mathbb{E}_\xi[U] = \mathbb{E}_\xi[\Delta U - \frac{1}{2}\|\nabla U\|^2 + R]$, and so

$$\boxed{d_{\text{eff}}^{\text{en}} - d_{\text{eff}}^{\text{den}} = -2\tau\, \mathbb{E}_\xi\big[R(x + \sqrt{\tau}\,\xi, \tau)\big].} \tag{8}$$

For the exact energy (Eq. 1), $R \equiv 0$ and the two forms coincide. That is Guth's theorem. For a learned U_θ they differ by exactly the averaged HJ residual. Guth et al. train U_θ with *dual* score matching: denoising score matching on $\nabla_y U$ plus time-score matching on $\partial_\tau U$ (their Eqs. 3, 5, 8). The second term penalises $\partial_\tau U$ mismatch, so it pushes R toward 0.

Related quantities. Integrating $d - d_{\text{eff}}$ over s gives the NLL (their Eq. 15): $-\log p(x) = \mathbb{E}_\xi[U(y, \tau_{\max})] - \int_0^{\tau_{\max}} \big(d - d_{\text{eff}}(x, \tau)\big)\frac{d\tau}{2\tau}$. Averaged over x, d_{eff} equals the MMSE dimension of Wu & Verdú. The $\tau \to 0$ limit is the local intrinsic dimension of the LIDL family.

2 B. MGD equations needed

All signs are *code* signs: $p_\theta \propto \exp(+\theta^\top \varphi)$, that is $\theta_{\text{code}} = -\theta_{\text{MGD paper}}$ (audit in `codes/utils_entropy.py`).

B.1 Ingredients

- Statistics $\varphi = (\varphi_1, \ldots, \varphi_r) : \mathbb{R}^d \to \mathbb{R}^r$, with analytic gradients $\nabla\varphi$. Code: the dict of potentials, each with `forward`, `grad(x, v)` $= \sum_i v_i \nabla\varphi_i(x)$, and `num_coefficients`. After `fit()` pruning, r is the fitted count (e.g. r=272, not 329).
- Cos interpolant, $\alpha_t = \pi t/2$: $I_t = \cos\alpha_t\, x_0 + \sin\alpha_t\, x_1$, with $x_0 \sim \mathcal{N}(0, I_d)$ and $x_1 \sim$ data.
- Moment path: $m_t = \mathbb{E}[\varphi(I_t)]$, and $\dot{m}_t = \frac{d}{dt} m_t$ (pathwise, `compute_rhs_dt_phi_I_t`).
- Gram matrix on an ensemble X: $G(X) = \frac{1}{N}\sum_n \nabla\varphi(X_n)\nabla\varphi(X_n)^\top$ ($r \times r$, `compute_G`).

B.2 The SDE step (`iteration_step_projection`)

Step k, with $h = t_{k+1} - t_k$ and noise level σ:

$$\text{predictor:}\quad G(x_k)\,\eta_k = \dot{m}_{t_k}, \qquad y_k = x_k + h\,\nabla\varphi(x_k)^\top \eta_k + \sqrt{2h}\,\sigma\,\xi_k, \tag{9}$$

$$\text{corrector:}\quad G(y_k)\,\theta_k^{\text{raw}} = m_{t_{k+1}} - \bar{\varphi}(y_k), \qquad x_{k+1} = y_k + \nabla\varphi(y_k)^\top \theta_k^{\text{raw}}, \qquad \theta_k = \theta_k^{\text{raw}}/(h\sigma^2). \tag{10}$$

The solves are Jacobi-scaled, with ridge `regularization` and a live mask (dead statistics set to 0). Noise plus corrector give $x_{k+1} = x_k + h\sigma^2\nabla\varphi^\top\theta_k + \sqrt{2h}\sigma\xi_k +$(transport). This is an Euler–Maruyama step of the Langevin dynamics $dX = \sigma^2\nabla\log p\,dt + \sqrt{2}\sigma\,dW$, and that dynamics leaves $p_{\theta_k} \propto e^{\theta_k^\top\varphi}$ invariant. The predictor drift moves the moments along $\dot{m}_t$.

B.3 Modelling assumption used by d_{eff}

MGD identifies the walker law at time t with the maximum-entropy exponential family that matches the moments:

$$p_t(z) \approx p_{\theta_t}(z) = \exp\big(\theta_t^\top \varphi(z) - \log Z_t\big), \qquad \mathbb{E}_{p_{\theta_t}}[\varphi] = m_t, \tag{11}$$

$$\Rightarrow \quad \nabla_z \log p_t(z) = \theta_t^\top \nabla\varphi(z). \tag{12}$$

z lives in interpolant coordinates ($z \sim I_t$, not in data units). Everything in Section 3 is evaluated in this model. The *true* law of I_t, $\mathcal{L}(I_t) = \mathcal{L}(\cos\alpha_t x_0 + \sin\alpha_t x_1)$, is *not* what is being measured.

B.4 Entropy and normaliser

From Eq. 11, $H(p_{\theta_t}) = \log Z_t - \theta_t^\top m_t$. If $\partial_t \log Z_t = \dot\theta_t^\top m_t$ (moments matched), then

$$\tfrac{d}{dt}H = -\theta_t^\top \dot m_t, \qquad \log Z_t = H(p_0) + \int_0^t (-\theta_u^\top \dot m_u)\, du + \theta_t^\top m_t, \qquad H(p_0) = \tfrac{d}{2}\log(2\pi e). \tag{13}$$

Code: `dH_k` $= -\theta_k^\top \dot m_{t_{k+1}}$ is saved per step as `dH_t_bound`. At $t = 1$ the integral is MGD's entropy bound (a lower bound on H, see `entropy_bound`), so Eq. 13 with the per-step θ_k is the pipeline's $\log Z_t$.

B.5 Time-regularised Θ_{reg} (what d_{eff} actually uses)

The per-step θ_k is noisy in t. `forward_regularised` stores, per coarse node, $M_k = G(y_k)$, $b_k = (m_{t_{k+1}} - \bar\varphi(y_k))/(h\sigma^2)$, $\Sigma_k = \mathrm{Cov}(\varphi(x_{k+1}))$ and $\dot m_k$. It then minimises (`reg_mode='moment'`, schedule `'uniform'`; `_solve_regularised_thomas`):

$$\mathcal{E}[\Theta] = \int_0^1 \Big[\tfrac12 \Theta^\top M_t \Theta - b_t^\top \Theta\Big] dt + \lambda \int_0^1 \Big[\tfrac12 \dot\Theta^\top \Sigma_t \dot\Theta - \dot m_t^\top \dot\Theta\Big] dt. \tag{14}$$

- First term: its pointwise minimiser is the corrector θ_k, i.e. the score-matching normal equations $M\Theta = b$.
- Second term: time-score matching for the family of Eq. 11. The model time score is $\partial_t \log p_\Theta = \dot\Theta^\top(\varphi - m)$ and the normal equations are $\Sigma_t \dot\Theta = \dot m_t$. This is the MGD analogue of Guth's ℓ_{TSM}, restricted to the span of φ.
- Discretised with trapezoid weights on t_{reg} (a block-tridiagonal SPD system, solved in float64 by block-Thomas). Options: `reg_ridge` (M_k += ridge $\cdot$ diagM_k) and `pin` (the $L_6 < 0$ bound). At $\lambda = 0$, $\Theta_{\text{reg}} = \theta_k$ (block-averaged).
- Output: `Theta_reg` of shape (n_{reg}, r) on grid t_{reg} (saved in the solution file for new runs; else rebuilt by `reconstruct_t_reg`). λ is chosen after the run (`resolve_theta_reg.py`).

3 C. Our implementation (`codes/effective_dimension.py`)

C.1 Map from MGD time to Guth noise

Rescale the interpolant: $y_t := I_t/\sin\alpha_t = x_1 + \cot\alpha_t\, x_0$. Given $x_1 = x$, $y_t \sim \mathcal{N}(x, \tau I_d)$ with

$$\tau(t) = \cot^2\alpha_t, \qquad s(t) = \tfrac12 \log\tau = \log\cos\alpha_t - \log\sin\alpha_t \tag{15}$$

(`tau_of_t`, `s_of_t`). τ decreases from ∞ at $t = 0$ to 0 at $t = 1$. So every t_{reg} node is one Guth scale. In particular $t = \frac12 \Leftrightarrow \tau = 1$, and the fine scales sit at $t \to 1$, where $\tau \approx (\pi(1-t)/2)^2$.

C.2 Model energy and score in Guth's coordinates

Change of variables $I_t = \sin\alpha_t\, y$, i.e. $q_\tau(y) = \sin^d\!\alpha_t\, p_t(\sin\alpha_t\, y)$. Write $z := \sin\alpha_t\, y$. With Eq. 11 and $\theta_t := \Theta_{\text{reg}}(t)$:

$$U_\theta(y,\tau) = -\theta_t^\top \varphi(z) + \log Z_t - d\log\sin\alpha_t, \tag{16}$$

$$\nabla_y \log q_\tau(y) = -\nabla_y U_\theta = \sin\alpha_t\; \theta_t^\top \nabla\varphi(z). \tag{17}$$

For $y = x + \sqrt{\tau}\,\xi$: $z = \sin\alpha_t\, x + \cos\alpha_t\, \xi$. This is exactly an interpolant sample with data endpoint x, so $\varphi, \nabla\varphi$ are evaluated in the range seen during the fit.

C.3 Denoiser form: `denoiser_d_eff`

Insert Eq. 17 into Eq. 4. Since $x - \mathbb{E}[x|y] = -\sqrt{\tau}\,\xi - \tau\nabla_y \log q_\tau$, the $1/\tau$ cancels:

$$d_{\text{eff}}^{\text{den}}(x,\tau) = \mathbb{E}_\xi\Big\| \xi + \sqrt{\tau}\,\sin\alpha_t\, \theta_t^\top \nabla\varphi(\sin\alpha_t x + \cos\alpha_t \xi)\Big\|^2 = \mathbb{E}_\xi\big\|\xi + \cos\alpha_t\, \theta_t^\top \nabla\varphi(z)\big\|^2, \tag{18}$$

using $\sqrt{\tau}\sin\alpha_t = \cos\alpha_t$. Algorithm, for each node i of the requested grid:

1. draw $\xi^{(1..n_{\text{mc}})} \sim \mathcal{N}(0, I_d)$ (fresh per node), and form $y = x + \sqrt{\tau_i}\,\xi$, $z = \sin\alpha_i\, y$;
2. $g = \sum_j \theta_{i,j} \nabla\varphi_j(z)$ via `grad_phi_projected` (float64 where the potential accepts it, else float32 upcast, remembered in `_NO_FLOAT64`);
3. return $\frac{1}{n_{\text{mc}}} \sum \|\xi + \sqrt{\tau_i}\sin\alpha_i\, g\|^2$, with shape (n_t, n_x).

Inputs: θ_t, $\nabla\varphi$. Not used: $\log Z$, m_t, or any derivative in τ. Unbiased for the model quantity at finite n_{mc}.

C.4 Energy form: `energy_d_eff`

1. $\log Z_t = H(t) + \theta_t^\top m_t$ at each node, with $H(t)$ from either
   - `h_bound_at`: trapezoid of the saved `dH_t_bound` (per-step θ_k) on the fine grid, interpolated to the nodes (the pipeline default, Eq. 13); or
   - `h_from_theta_reg`: $H(p_0) - \int \Theta_{\text{reg}}^\top dm$ on the t_{reg} grid. This makes $\partial_t \log Z = \dot\Theta^\top m$ exact; it is not covered by MGD's bound proof.

   m_t comes from the saved `aux_moments` (`barphi_e`) or from `recompute_m_t` (data and schedule only).
2. $\bar U_i(x) = \frac{1}{n_{\text{mc}}} \sum_\xi U_\theta(x + \sqrt{\tau_i}\xi, \tau_i)$ from Eq. 16 (`phi_stack`, float64). By default the same ξ draws are used at every node (`common_noise=True`), so the MC noise is smooth in s.
3. $\partial_s \bar U$ by a local weighted quadratic fit over $2h{+}1$ neighbouring nodes (`local_poly_derivative`, default $h = 5$; the window is shifted, not shrunk, at the ends). Return $d - \partial_s \bar U$.

Inputs: θ_t, φ, $\log Z_t$ (hence m_t and H). Only the s-dependence of $\log Z_t$ matters: a constant offset in H cancels, while an error $\delta\log Z_t$ shifts every x by $-\partial_s \delta\log Z_t$.

C.5 What the gap measures

By Eq. 8 applied to U_θ of Eq. 16, at equal θ_t, points and noise:

$$d_{\text{eff}}^{\text{en}} - d_{\text{eff}}^{\text{den}} = -2\tau\, \mathbb{E}_\xi\big[R_\theta(y,\tau)\big] \;+\; (\text{MC noise}) \;+\; (\text{error of } \partial_s \text{ on the grid}), \tag{19}$$

where R_θ is the HJ residual (Eq. 7) of the MGD model energy, written out in terms of θ_t, φ in Eq. 32. It is nonzero whenever the family $\{p_{\theta_t}\}$ with this $\log Z_t$ is not a heat flow in y. Three sources are folded into R_θ: (a) the exponential family is moment-matched, not score-matched (Eq. 11); (b) Θ_{reg} imposes time-score consistency only through the λ-term of Eq. 14, and only inside span φ; (c) $\partial_t \log Z_t$ from `h_bound_at` contains $(\Theta_{\text{reg}} - \theta_k)^\top \dot{m}$, the mismatch between the two θ paths. The denoiser form is free of (c) and of the ∂_s numerics, and needs no heat-flow assumption. Hence it is the reported estimator (Section 11).

C.6 How it is run (`deff_diagnostics/deff_latest_run_check.py`)

- Load the run's `fitted_potentials/` (`load_fitted_potentials`). Without them r is wrong and every θ index is shifted.
- Load Θ_{reg} and t_{reg} from the solution file. Check that $t_{\text{reg}} \subset t[1{:}]$ (assert) and compare with `reconstruct_t_reg`. Take $m_t =$ `barphi_e` at those exact nodes.
- Evaluation points x: rows of the run's own normalised training data x_1, split into typical/rare by the 95th percentile of max |lag-1 increment| (`extract_rare_typical`). Guth evaluate on held-out images; here the points are in the fit set.
- Evaluation nodes: a subset of t_{reg} chosen uniformly in s. Denoiser at two n_{mc} values (default 64, 256) in float64, plus float32 as a precision check; energy at $\{\text{bound}, \Theta_{\text{reg}}\} \log Z \times \{\text{fresh}, \text{common}\}$ noise. Per-seed CSVs are written and then plotted in `turbulence/effective_dimension_deff_final.ipynb`.

C.7 Correspondence table

Guth et al.	MGD / this code	function
noise variance t (here τ)	$\cot^2(\pi t/2)$	`tau_of_t`
$s = \frac{1}{2}\log\tau$	$\log\cot(\pi t/2)$	`s_of_t`
$y = x + \sqrt{\tau}\xi$	$z = \sin\alpha_t x + \cos\alpha_t \xi = \sin\alpha_t\, y$	(inline)
$-\nabla_y U_\theta(y,\tau)$	$\sin\alpha_t\, \Theta_{\text{reg}}(t)^\top \nabla\varphi(z)$	`grad_phi_projected`
$U_\theta(y,\tau)$	$-\Theta_{\text{reg}}^\top \varphi(z) + \log Z_t - d\log\sin\alpha_t$	`phi_stack`
$\log Z$ normalisation	$H(t) + \Theta_{\text{reg}}^\top m_t$	`h_bound_at`/`h_from_theta_reg`
learned energy network	exponential family on φ, θ from MGD	`Theta_reg`
DSM loss	$\frac{1}{2}\Theta^\top M\Theta - b^\top\Theta$ (corrector)	`compute_theta`
TSM loss	$\lambda(\frac{1}{2}\dot\Theta^\top\Sigma\dot\Theta - \dot m^\top\dot\Theta)$	`_solve_regularised_thomas`
Eq. 11	Eq. 18	`denoiser_d_eff`
Eq. 12	$d - \partial_s \mathbb{E}_\xi U_\theta$	`energy_d_eff`

Part II: derivation and debugging record

4 Guth's d_{eff}: recap

Let q_τ be the law of $y = x + \sqrt{\tau}\,\xi$, $\xi \sim \mathcal{N}(0, I_d)$, x drawn from the data. Two equivalent definitions:

$$\text{energy form:} \qquad d_{\text{eff}}(x,\tau) = d - \partial_s \, \mathbb{E}_y\big[U(y,\tau) \mid x\big], \qquad s = \tfrac{1}{2}\log\tau, \tag{20}$$

$$\text{denoiser form:} \qquad d_{\text{eff}}(x,\tau) = \frac{1}{\tau}\,\mathbb{E}_y\big[\|x - \mathbb{E}[x \mid y]\|^2 \mid x\big], \tag{21}$$

where $U(\cdot,\tau) = -\log q_\tau(\cdot)$ and, by Tweedie's formula, $\mathbb{E}[x \mid y] = y + \tau\,\nabla_y \log q_\tau(y)$. Both reduce to d for an isotropic Gaussian and to (less than) d when the data lies near a lower-dimensional manifold, at scales τ finer than the manifold's local curvature.

5 MGD's schedule and sign convention

MGD's Cos stochastic interpolant (`sde_routines.py`, `compute_ interpolant`, `interpolant='Cos'` – the only schedule `forward_regularised` supports) is

$$I_t = \cos(\alpha_t)\, x_0 + \sin(\alpha_t)\, x_1, \qquad \alpha_t = \tfrac{\pi t}{2}, \qquad x_0 \sim \mathcal{N}(0, I_d),\ x_1 = x \text{ (data)}. \tag{22}$$

Sign convention. This codebase fits θ_t for $p_\theta(x) \propto \exp(+\theta^\top \varphi(x))$ – the *opposite* of the MGD paper's $\exp(-\theta^\top\varphi(x))$. This is audited directly against the code in `codes/utils_entropy.py` ("Sign convention audit"): the corrector step in `iteration_step_projection` is $x_{k+1} = y_k + \nabla\varphi^\top\theta_{\text{raw}}$ (a *plus*), giving a continuous-time drift $\sigma^2\nabla\varphi^\top\theta_t$ that must equal $\sigma^2\nabla\log p_t$ for the transport to preserve p_t, so

$$\nabla_x \log p_t(x) \;=\; +\theta_t^\top \nabla\varphi(x). \tag{23}$$

Independently corroborated by $dH_k = -\theta_k^\top \dot{m}_t$ (the entropy increment already computed in `iteration_step_projection`) and by $\log Z_{\theta_1} \gtrsim H_{\text{bound}} + \theta_1^\top m_1$ (MGD Eq. 8/30 as implemented in `utils_entropy.log_Z_bound`). Every formula below is re-derived directly in this (+) convention – not by substituting $\theta_{\text{paper}} = -\theta_{\text{code}}$ into the paper's stated formulas.

6 The VE rescaling

Define $y_t := I_t/\sin(\alpha_t) = x_1 + \cot(\alpha_t)\, x_0$. Conditional on $x_1 = x$,

$$y_t \mid x \;\sim\; \mathcal{N}\big(x,\ \tau(t)\, I_d\big), \qquad \tau(t) = \cot^2(\alpha_t), \tag{24}$$

monotonically decreasing from $\tau(0) = \infty$ to $\tau(1) = 0$. So MGD's Cos schedule already sweeps every VE noise scale, and t_{reg} (the grid θ_t is fit on) IS a τ-grid via Eq. 24.

Write q_τ (Guth's VE marginal, density in y) in terms of p_t (MGD's marginal, density in I_t) by the change of variables $I_t = \sin(\alpha_t)\, y$:

$$q_\tau(y) = \sin^d(\alpha_t)\ p_t\big(\sin(\alpha_t)\, y\big). \tag{25}$$

Differentiating and using Eq. 23 (evaluated at $I_t = \sin(\alpha_t)y$):

$$\nabla_y \log q_\tau(y) \;=\; \sin(\alpha_t)\,\nabla_x \log p_t(\sin(\alpha_t)y) \;=\; +\sin(\alpha_t)\,\theta_t^\top \nabla\varphi\big(\sin(\alpha_t)\, y\big). \tag{26}$$

The extra $\sin(\alpha_t)$ factor is the chain-rule term the derivation asked to double-check – confirmed, with the (+) sign from Eq. 23 (a paper-convention derivation would carry a minus here).

Practical note. $\sin(\alpha_t)\, y = \sin(\alpha_t)\, x + \cos(\alpha_t)\,\xi$ for $y = x + \sqrt{\tau}\,\xi$ (since $\sqrt{\tau} = \cot\alpha_t$) – literally `compute_interpolant`'s own Cos schedule with x as a single-point data endpoint and ξ as fresh noise. So every $\varphi/\nabla\varphi$ evaluation below stays inside the numerical range the fit itself already explores; nothing is evaluated off-distribution in an uncontrolled way.

7 Denoiser form

Substituting Eq. 26 into Tweedie's formula, $x - \mathbb{E}[x \mid y] = -\sqrt{\tau}\,\xi - \tau\,\nabla_y \log q_\tau(y)$, so the $1/\tau$ in Eq. 21 cancels analytically:

$$\boxed{d_{\text{eff}}(x,\tau) \;=\; \mathbb{E}_\xi\Big[\;\big\|\,\xi + \sqrt{\tau}\,\sin(\alpha_t)\,\theta_t^\top \nabla\varphi\big(\sin(\alpha_t)\,y\big)\,\big\|^2\;\Big], \qquad y = x + \sqrt{\tau}\,\xi.} \tag{27}$$

No $\log Z_t$, no m_t, no normalisation constant – only θ_t and $\nabla\varphi$. This is the primary estimator (`effective_dimension.denoiser_d_eff`).

8 Energy form

From $U(y,\tau) = -\log q_\tau(y)$,

$$U(y,\tau) \;=\; -\theta_t^\top \varphi\big(\sin(\alpha_t) y\big) \;+\; \log Z_t \;-\; d\log\sin(\alpha_t). \tag{28}$$

The last term is analytic in t; the first two are not, so Eq. 20's ∂_s is taken numerically, by finite-differencing $\mathbb{E}_y[U(y,\tau) \mid x]$ directly across the t_{reg} grid (converted to $s = \frac{1}{2}\log\tau(t)$), rather than splitting analytic/Monte-Carlo pieces (`effective_dimension.energy_d_eff`).

$\log Z_t$ **in the pipeline.** Our pipeline supplies $\log Z_t$ via MGD's Prop. 4.3/Eq. 30 construction,

$$\log Z_t \;=\; H_{\text{bound}}(t) + \theta_t^\top m_t, \qquad H_{\text{bound}}(t) = H(p_0) + \int_0^t dH_s\,ds, \tag{29}$$

generalised from the paper's stated $t = 1$ endpoint to any t (the integral is just truncated earlier). $H_{\text{bound}}(t)$ is built by trapezoid-integrating the already-saved dH_t^{bound} (`effective_dimension.h_bound_at`); m_t is either the saved `aux_moments` (the run's own realised target moments) or, if not pulled from the cluster yet, recomputed from the data and schedule, $m_t = \mathbb{E}_Z[\varphi(\cos\alpha_t\, Z + \sin\alpha_t\, x_1)]$ – independent of any saved particle trajectory, so no SDE rerun either way (`effective_dimension.recompute_m_t`). An alternative, θ_t-only construction $\log Z_t = -H(p_0) - \int_0^t \theta_s^\top \dot{m}_s\,ds$ is also implemented (`effective_dimension.h_from_theta_reg`); the two choices differ by less than 0.3 nats on the production runs we have measured (an x-independent shift $-d(\delta\log Z)/ds$ of the energy-form curve), so neither is the explanation for the much larger disagreement we discuss next.

9 The energy–denoiser gap: a direct read-out of model residual

Why Eqs. 20 and 21 agree in Guth's setting. Their equivalence is proved (their Appendix B.3) by one integration-by-parts step that uses the Fokker–Planck equation of the noising:

$$\partial_\tau\, q_\tau(y) \;=\; \tfrac{1}{2}\Delta_y\, q_\tau(y). \tag{30}$$

For a density q_τ obtained by heat-diffusing a single underlying q_0, Eq. 30 holds exactly and the two definitions coincide analytically.

Why they do *not* agree for MGD. Our pipeline does not produce a density that solves Eq. 30 along τ. At each τ we solve a dual problem that fixes a few averages. In interpolant coordinates $z = \sin\alpha_t\, y$:

$$p_{\theta_t}(z) \;\propto\; \exp\big(+\theta_t^\top \varphi(z)\big), \qquad \mathbb{E}_{p_{\theta_t}}[\varphi] \;=\; m_t, \qquad q_\tau(y) = \sin^d\!\alpha_t \; p_{\theta_t}(\sin\alpha_t\, y). \tag{31}$$

That is, the family $\{p_{\theta_t}\}$ is *moment-matched* to the true noisy data, not *score-matched*. The two conditions are related but strictly inequivalent: moment matching constrains a finite number of integrals $\int \varphi_j\, p_\theta$; score matching asks that $\nabla_y \log p_\theta$ agree with $\nabla_y \log q_\tau$ pointwise in y, which is much stronger. A diffusion-model score network trained to convergence satisfies the second (modulo Fisher-divergence training loss); MGD only satisfies the first (modulo moment-residual).

The residual is the Hamilton–Jacobi residual of Part I, $R = \partial_\tau U - \frac{1}{2}\Delta_y U + \frac{1}{2}\|\nabla_y U\|^2$ (Eq. 7). It enters the gap exactly as $d_{\text{eff}}^{\text{en}} - d_{\text{eff}}^{\text{den}} = -2\tau\, \mathbb{E}_\xi[R]$ (Eq. 8). Take U from Eq. 28, i.e. $U = -\theta_t^\top \varphi(z) + \log Z_t - d\log\sin\alpha_t$ with $z = \sin\alpha_t\, y$. Use $\partial_\tau|_y = (dt/d\tau)\,\partial_t|_y$, $dt/d\tau = -\sin^3\!\alpha_t/(2\dot\alpha\cos\alpha_t)$, $\dot\alpha = \pi/2$, $\nabla_y = \sin\alpha_t \nabla_z$ and $\partial_t z|_y = \dot\alpha\cot\alpha_t\, z$. Write $\dot\theta_t = d\theta_t/dt$ and $\delta_t := \partial_t \log Z_t - \dot\theta_t^\top m_t$ ($\delta_t = 0$ when $\log Z_t$ is consistent with the moments). Then

$$\begin{aligned} R(y,\tau) \;=\; & \frac{\sin^3\alpha_t}{2\dot\alpha\cos\alpha_t}\Big[\; \underbrace{\dot\theta_t^\top\big(\varphi(z) - m_t\big)}_{\text{(i) rate of change of }\theta} \;-\; \underbrace{\delta_t}_{\text{(v) } \log Z \text{ drift}} \;\Big] \\ & + \tfrac{1}{2}\sin^2\alpha_t\Big[\; \underbrace{\theta_t^\top \Delta_z \varphi(z)}_{\text{(ii) basis curvature}} \;+\; \underbrace{\big\|\theta_t^\top \nabla_z \varphi(z)\big\|^2}_{\text{(iii) score magnitude}} \;+\; \underbrace{z\cdot\theta_t^\top \nabla_z\varphi(z) + d}_{\text{(iv) rescaling } I_t = \sin\alpha_t\, y} \;\Big]. \end{aligned} \tag{32}$$

Term (i) measures whether the Lagrange multiplier moves in step with the moments at the point z. Term (ii) is the Laplacian of the basis functions weighted by their multipliers, a curvature that moment matching never constrains. Term (iii) is the squared score. Term (iv) comes from the $\sin\alpha_t$ rescaling between MGD's I_t and Guth's y, through both the y-dependence of z and the Jacobian $-d\log\sin\alpha_t$. Term (v) is nonzero when $\log Z_t$ is not built from the same θ path (e.g. `h_bound_at` with per-step θ_k while $\theta_t = \Theta_{\text{reg}}$; then $\delta_t = (\Theta_{\text{reg}} - \theta_k)^\top \dot m_t$). *Check.* Take $\varphi(z) = -\frac{1}{2}\|z\|^2$, $\theta_t = 1/v_t$ and $v_t = \cos^2\alpha_t + \sigma_{\text{data}}^2 \sin^2\alpha_t$ (the exact Gaussian case of Section 10). Then the first bracket gives $\frac{\sin^4\alpha_t(\sigma_{\text{data}}^2 - 1)}{2v_t^2}(\|z\|^2 - dv_t)$ and the second gives its negative, so $R \equiv 0$. For a true heat-equation solution the terms always cancel. For the moment-matched family they generically do not.

What is measured on the production runs. The energy and denoiser forms (Eqs. 20 and 21) use the *same* θ_t, *same* φ and *same* evaluation grid; they can be compared point-for-point. On the final turbulence runs (d=256, λ=5×10^{-4}, 5 seeds, with the x-dependent Monte-Carlo noise cancelled by common random numbers across τ-nodes), the mean gap $d_{\text{eff}}^{\text{energy}} - d_{\text{eff}}^{\text{denoiser}}$ is

- at most a few units at the extremes $\tau \to 0$ and $\tau \to \infty$, where both forms either saturate at d or vanish;
- of order -50 to -70 in the mid-τ range $\tau \in [10^{-4}, 10^{-2}]$, with per-point standard deviation ~ 200;
- reproducible across arms (same signal with λ=10^{-3} constrained L_6<0 *and* λ=5×10^{-4} unconstrained),

exactly tracking the location of the physical burst scale in the data (the bump in $d_{\text{eff}}^{\text{denoiser}}$ sits at the same τ). The two $\log Z_t$ constructions differ by ≤ 0.3 nats in $\log Z_t$ itself. They shift $d_{\text{eff}}^{\text{energy}}$

by $-\partial_s$ of that difference (term (v) above), which is negligible next to a gap of tens of units of d_{eff}. The remainder is the integrated residual $-2\tau\,\mathbb{E}_\xi[R]$ of Eq. 32, and it is *structurally present* in any moment-matched family: it is the quantitative cost of substituting a dual on $\theta \cdot m$ for the stronger Fisher-divergence objective a score network would minimise.

Physical reading. The gap is largest exactly where our denoiser-form d_{eff} shows a distinct bump above its monotone trend – at the burst scale $\tau \sim 10^{-3}$, in the mid-τ window where the noising transport actively exchanges information with the data manifold. There, several moment constraints are active simultaneously (second-order moments, scattering coefficients across adjacent scales, the L_6 intermittency term), the Lagrange multipliers change fastest, and Eq. 32's terms (i) and (iii) peak together. At both extremes the family trivialises: $p_\theta(\cdot, \tau)$ is nearly Gaussian in y-coordinates at large τ (one moment active) and nearly a delta at the data at small τ, and in both limits the contributions in Eq. 32 cancel to within a few units. The gap is therefore not a numerical artefact; it is the finger-print of how moment-matched p_θ differs from the exact noisy marginal, with amplitude controlled by the physics of the transport.

10 Validation against a closed-form Gaussian test

To separate genuine implementation bugs from the expected bound bias above, both estimators were checked against an exact, independently-derived ground truth – no SDE, no fitting, no θ_t from any real run. For $x \sim \mathcal{N}(0, \Sigma)$, $y = x + \sqrt{\tau}\,\xi$, $q_\tau = \mathcal{N}(0, \Sigma + \tau I)$ exactly, so Tweedie's formula gives (diagonal basis, $\Sigma = \text{diag}(\sigma_i^2)$):

$$d_{\text{eff}}(x, \tau) \;=\; \sum_i \frac{\tau x_i^2 + \sigma_i^4}{(\sigma_i^2 + \tau)^2}, \tag{33}$$

verified against brute-force $(\Sigma + \tau I)^{-1}$ linear algebra (agrees to $\sim 10^{-16}$) and its $\tau \to 0$ ($\to d$) and isotropic-expectation ($\to d\sigma^2/(\sigma^2 + \tau)$) limits. For *this* synthetic data MGD's own p_t is also exactly Gaussian, so $\theta_t = -d/(2v(t))$ and $\log Z_t = \frac{d}{2}\log(2\pi v(t))$, $v(t) = \cos^2\alpha_t + \sin^2\alpha_t\,\sigma^2_{\text{data}}$, are exact too (single-coefficient potential $\varphi(x) = \overline{x^2}$) – feeding these directly into the *unmodified* `denoiser_d_eff`/`energy_d_eff` isolates implementation bugs with zero fitting noise and zero bound bias.

Using the real production grid (d=256, t_{reg} reconstructed from `nt=40000`, `n_subsample=200`): **denoiser_d_eff matched the oracle to $< 0.5\%$ relative error everywhere** (10 decades of τ, $|\theta|$ up to 512) – its implementation is correct. `energy_d_eff`, despite the exact inputs, showed errors up to 1320%, concentrated at the *coarsest* end of the grid.

10.1 Root cause: $s = \frac{1}{2}\log\tau$ is highly non-uniform, and it was differentiated naively

$\tau(t) = \cot^2(\alpha_t)$ diverges (logarithmically, in s) at both $t \to 0$ and $t \to 1$, so $s(t)$ is intrinsically badly stretched near the ends of the range *regardless of the t-schedule* – confirmed directly: a perfectly *uniform*-in-t grid has an even worse max/min local-Δs ratio (340×) than the actual Cos-schedule production grid (67×, since bunching points near $t = 1$ partially compensates). Differentiating $\mathbb{E}_y[U]$ across this grid with a raw finite difference (`numpy.gradient`, which defaults to `edge_order=1`) has two compounding failure modes, both confirmed in isolation:

(A) **Boundary truncation bias.** `edge_order=1` is only first-order accurate at the array's first/last nodes – on a noiseless synthetic quadratic in s, this alone produces a 6.4% error at the boundary (exact at every interior node).

(B) **Monte-Carlo noise amplification** (the dominant effect): $U(t)$ is itself MC-estimated, and dividing its noise by a small local Δs amplifies it. An n_{mc} sweep at the worst node showed the error shrinking monotonically as n_{mc} grew ($0.77 \to 0.29 \to 0.23$ over 4k→16k→32k), confirming noise, not just truncation.

The natural first fix, `edge_order=2`, removes (A) exactly (boundary error drops to machine precision on the noiseless check) but **does not help, and can make the realistic case worse** (max relative error 17.9 vs. 10.4) – its wider stencil has larger coefficients under uneven spacing, amplifying (B) more than it fixes (A).

10.2 Fix: a local polynomial derivative

`energy_d_eff` now differentiates $\mathbb{E}_y[U]$ via `effective_dimension.local_poly_derivative`: at each node, a weighted-least-squares quadratic is fit through a window of $2h+1$ *neighbouring indices* (shifted, not shrunk, at the array edges, so every node – including the boundary – gets a full window) and differentiated analytically. This fixes (A) exactly (machine precision on the noiseless check, any window size) and reduces (B) by averaging over several points rather than two. Validated on the Gaussian test ($h = 5$, the shipped default): overall median relative error dropped $\sim 6\times$ ($1.4\% \to 0.22\%$), and the fraction of the grid accurate to $< 1\%$ roughly doubled ($39\% \to 70\%$ of nodes, with the "becomes reliable" point moving from $t \approx 0.80$ down to $t \approx 0.41$). The few nodes at the very coarsest end of the grid remain the least reliable part of an energy-form curve on principle – no local model, however fit, is well-conditioned there – and should be read with that in mind; the denoiser form has no such issue, since it involves no differentiation step at all.

11 Which form to use in the paper

Recommendation. Use the denoiser form (Eq. 21, implemented in `denoiser_d_eff`) as the primary estimator. The energy form (Eq. 20) belongs in an appendix or methods section as a diagnostic, not in a main figure. Three reasons:

1. *Unbiased for the quantity we care about.* What we want to report is a well-defined property of our *learned* model p_θ: how much of x remains after one scale of noising, averaged over noise draws. That is literally Eq. 21, and it uses only the model's score $+\sin\alpha_t\,\theta_t^\top \nabla\varphi$ via Tweedie's formula – no assumption about whether $p_\theta(\cdot,\tau)$ is the exact noisy marginal of anything. The energy form (Eq. 20) silently assumes p_θ satisfies Eq. 30, which MGD's moment-matched family does not (Section 9). Reporting the energy form means reporting a quantity biased by the model's FPE residual, with a mandatory footnote explaining that the bias is physical rather than numerical.

2. *Closer to the MGD construction.* MGD is defined by moment matching and generates samples by running an SDE whose drift is the model score. The denoiser form reads that same score through Tweedie's formula and measures how well it inpaints noisy inputs – i.e. it measures what the model is designed to do. The energy form additionally requires the normaliser $\log Z_\tau$, a derived quantity that MGD builds (Eq. 29) but never places constraints on, and whose definition is slightly implementation-dependent (bound vs. θ_t-only, see Section 8). The denoiser form has no $\log Z$ dependence at all.

3. *Fewer numerics-side worries.* The energy form takes a numerical derivative $\partial_s \mathbb{E}_y[U|x]$ across the t_{reg} grid; this amplifies Monte-Carlo noise unless common random numbers are used across

τ-nodes (Section 10's controlled test showed this was the dominant implementation hazard, and the mitigation changed defaults). The denoiser form has no differentiation step at all. Its one real downside – larger seed-to-seed scatter at $\tau \gg 1$, where d_{eff} itself is small – is cured by seed averaging; a 30-seed production run brings the floor well below the physical mean value at every τ.

Suggested paper phrasing. "*For a sample x and noise scale τ, we report the local effective dimensionality*

$$d_{\text{eff}}(x,\tau) \;=\; \tfrac{1}{\tau}\,\mathbb{E}_{\xi}\big[\,\|x - \mathbb{E}[x \mid y]\|^2 \;\mid\; x\big]\,, \qquad y = x + \sqrt{\tau}\,\xi,$$

*with the posterior mean given by Tweedie's formula applied to the model's own score, $\nabla_y \log p_\theta(y,\tau) = +\sin\alpha_t\,\theta_t^\top \nabla\varphi(\sin\alpha_t\, y)$. This is Guth, Kadkhodaie & Simoncelli's [*arXiv:2506.05310*] Eq. 11; the equivalent Eq. 12 (an energy-based definition requiring Fokker–Planck for p_θ) differs from Eq. 11 by a residual quantifiable in the moment-matched setting and reported in Appendix X as a diagnostic of model mismatch.*"

The appendix cross-reference, built from Section 9, documents the gap as the integrated FPE residual $R(y,\tau)$ of Eq. 32 and is one short figure showing the gap curve and its concentration at the burst scale. That removes any ambiguity for a reader comparing MGD's d_{eff} to a diffusion-model baseline whose two forms would agree.

12 Dictionary

Guth et al.	MGD (this codebase)
VE noise scale τ	$\tau(t) = \cot^2(\alpha_t)$, $\alpha_t = \pi t/2$
score $\nabla_y \log q_\tau(y)$	$+\sin(\alpha_t)\,\theta_t^\top \nabla\varphi(\sin(\alpha_t)y)$
energy $U(y,\tau)$	$-\theta_t^\top \varphi(\sin(\alpha_t)y) + \log Z_t - d\log\sin(\alpha_t)$
$\log Z_\tau$	$H_{\text{bound}}(t) + \theta_t^\top m_t$ *(bound, Eq. 29)*
noised sample $y = x + \sqrt{\tau}\,\xi$	$\sin(\alpha_t)\,y =$ `compute_interpolant`(x,ξ)
per-τ fit	$\theta_t \equiv$ `Theta_reg[i]`, grid t_{reg}

13 What's used from a saved run

- `Theta_reg`, shape (n_t, r) – the regularised θ_t trajectory (`codes/regularised_theta.py`); used exclusively (not the raw, finer θ_t) for a single self-consistent grid and to avoid a pre-fix off-by-one alignment issue present in some saved runs' raw θ_t/`dH_t_bound` pairing.
- t_{reg} – `Theta_reg`'s own coarse time grid; not saved for the d=256 runs on disk as of this writing, so it is reconstructed offline (`reconstruct_t_reg`) by replaying `SDE.forward_regularised`'s block-accumulation and near-node-pruning exactly, from the saved fine grid t alone – verified to reproduce the on-disk `Theta_reg` length exactly.
- Each run's `fitted_potentials/` – $\varphi, \nabla\varphi$ with the SAME pruned active-statistic set the fit used (some potentials auto-prune during `fit()`; skipping this step silently mis-indexes every θ coefficient).
- `dH_t_bound` + the fine grid t – integrated to $H_{\text{bound}}(t)$ (Eq. 29), needed only by the energy form.

- `aux_moments` (`barphi_e`) if pulled from the cluster, else m_t recomputed from the run's own x_1 – needed only by the energy form.

14 Caveats

- **The energy–denoiser gap is model residual, not an implementation defect** (Section 9). The earlier working hypothesis – that the gap was "bound bias" in $\log Z_t$ – was ruled out on 2026-10-13/14: a Gaussian oracle test with exact inputs (Section 10) reproduces the energy form to the Monte-Carlo noise floor, and on the production runs the gap persists at -50 to -70 (units of d_{eff}, which is dimensionless) across two independent regularisation arms (λ=10^{-3} constrained L_6<0 and λ=5×10^{-4} unconstrained) while the two alternative $\log Z_t$ constructions differ by ≤ 0.3 nats in $\log Z_t$. The gap is the integrated residual $-2\tau\,\mathbb{E}_\xi[R]$ of Eq. 32, concentrated at the burst scale $\tau \sim 10^{-3}$.

- θ_t becomes numerically extreme as $t \to 1$ ($\tau \to 0$): every seed inspected has one dominant, near-null-direction coefficient at the finest scale, with $|\theta|$ up to $\sim 10^8$ on the final runs. This is harmless for the actual energy/score (both stay smooth off-support – checked directly), but $\theta_t^\top \varphi$ and $\theta_t^\top \nabla\varphi$ must be computed in double precision, not the float32 the values are stored in. The current defaults (`denoiser_d_eff(float64=True)`, `energy_d_eff(float64=True)`) do this automatically, with a silent fall-back to float32 for the handful of potentials whose internal buffers do not accept float64 (`Scattering_Fourth_Order_Mod2_*`, memoised per potential in `_NO_FLOAT64`). Measured $|\text{den32} - \text{den64}|$ on the final runs stays at $\sim 10^{-4}$ while the Monte-Carlo noise is ~ 1 – precision is not currently a bottleneck.

- The finest available τ is set by how close to $t = 1$ the run's schedule and `n_subsample` block reach, not by an arbitrary choice – see `codes/effective_dimension.py`'s `reconstruct_t_reg` for the exact grid actually available for a given run.

Implementation changelog

- *2026-09-23:* `denoiser_d_eff`/`energy_d_eff` cast θ_t to `x.dtype` for `potentials.grad()` but not to `x.device` – harmless on a single-device (CPU) run, but a hard crash on Jean Zay, where θ_t/m_t load onto CPU (`map_location='cpu'`) while `x`/the potentials live on `cuda`. Fixed: both are now moved to `x.device` inside each function.

- *2026-09-24:* the differentiation bug and fix described in Section 10 (`local_poly_derivative` replacing a raw `numpy.gradient` call in `energy_d_eff`).

- *2026-10-13:* `energy_d_eff` and `denoiser_d_eff` default to `float64=True`; the energy form additionally defaults to `common_noise=True` (same Gaussian noise draw ξ at every τ-node, so the pointwise $\partial_s \mathbb{E}_y[U|x]$ differentiation does not amplify Monte-Carlo noise). A helper `_call_maybe_f64` inside `effective_dimension.py` remembers per-potential which basis classes refuse float64 input (notably `Scattering_Fourth_Order_Mod2_Real/Imag_Q1`, whose internal buffers are float32 only), and falls back to float32 for those while keeping the rest in float64. New helpers `phi_stack` and `h_from_theta_reg` added; the latter is the θ_t-only alternative $\log Z_t$ construction discussed in Section 8.

- *2026-10-14:* the earlier working hypothesis that the energy–denoiser gap was "bound bias" in $\log Z_t$ (previous version of Section 5) was ruled out and replaced with the FPE-residual

explanation of Section 9: the gap persists at the same -50 to -70 level (units of d_{eff}) across two independent regularisation arms with different λ and different L_6 sign-constraint choices, while the two possible $\log Z_t$ constructions differ by ≤ 0.3 nats in $\log Z_t$. The notes' recommendation to use the denoiser form as the primary estimator (Section 11) is a consequence of this. Diagnostic script: `deff_diagnostics/deff_latest_run_check.py` (per-seed CSVs); CSV-loader notebook: `turbulence/effective_dimension_deff_final.ipynb`.

- *2026-10-15:* Part I added. Eq. 32 corrected. The previous version had the sign of the score-magnitude term inconsistent with the other two terms. It also wrote the family in y instead of $z = \sin\alpha_t y$, so it lacked the rescaling term (iv) and the $dt/d\tau$ factor. The new version is checked to vanish identically in the Gaussian case. The gap is expressed in units of d_{eff} (dimensionless), not nats. The stale "lower-bound bias" wording in the `energy_d_eff` docstring is replaced.